# Computational Physics Homework 7 Report

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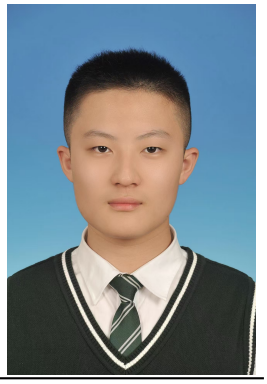

This report summarizes the four homework tasks shown on the last pages of *Computational Physics Lecture 7*. The folder contains several drafts and executable files, so I first identified the versions that are most likely to be the final submissions, and then organized their mathematical results and numerical outputs into one English report.

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# 1 Selected Source Files

The working directory contains several versions of the homework. The files chosen as the most likely final drafts are listed in Table 1. The main criterion is whether a file matches the lecture task directly and contains the complete required workflow instead of only a partial experiment.

表 1: Selected final drafts used in this report.

| Task | Selected file(s) | Reason |
|---|---|---|
| Problem 1 | `hw7_1/main.tex`, `hw7_1/main.pdf` | Clean electronic version of the handwritten derivation. The file `手写稿.pdf` is treated as the original handwritten draft rather than the final submission. |
| Problem 2 | `hw7_2/main.py` | Covers the full assignment: table-data approximation, error sweep for many $h$ values, and figure generation. The C file only covers the second part. |
| Problem 3 | `hw7_3/main.py` | Only complete implementation in the folder, including both the table-data test and the $h$-dependence study. |
| Problem 4(a) | `hw7_4/a/a.c` | Direct implementation of the first derivative program requested in the lecture. |
| Problem 4(b) | `hw7_4/b/b.c` | Direct implementation of the multi-$\delta$ accuracy test. |
| Problem 4(c) | `hw7_4/c/c.py` | Matches the lecture prompt most directly by reproducing the forward-difference error plot for $f(x) = \sin x$. The files `c+.py` and `c++.py` appear to be later extensions to higher-order formulas. |

# 2 Theoretical Analysis

## 2.1 Problem 1: finite-difference formulas

Starting from the shift operator $Ef(x) = f(x+h)$, the forward, backward, and central differences are

$$\Delta = E - 1, \qquad \nabla = 1 - E^{-1}, \qquad \delta = E^{1/2} - E^{-1/2}.$$

Expanding these operators and combining them with Taylor series gives the derivative formulas below.

### 2.1.1 Forward differences

$$f''(x) \approx \frac{f(x+2h) - 2f(x+h) + f(x)}{h^2}, \qquad O(h).$$

$$f^{(3)}(x) \approx \frac{f(x+3h) - 3f(x+2h) + 3f(x+h) - f(x)}{h^3}, \qquad O(h).$$

$$f^{(4)}(x) \approx \frac{f(x+4h) - 4f(x+3h) + 6f(x+2h) - 4f(x+h) + f(x)}{h^4}, \qquad O(h).$$

$$f^{(5)}(x) \approx \frac{f(x+5h) - 5f(x+4h) + 10f(x+3h) - 10f(x+2h) + 5f(x+h) - f(x)}{h^5}, \qquad O(h).$$

### 2.1.2 Backward differences

$$f''(x) \approx \frac{f(x) - 2f(x-h) + f(x-2h)}{h^2}, \qquad O(h).$$

$$f^{(3)}(x) \approx \frac{f(x) - 3f(x-h) + 3f(x-2h) - f(x-3h)}{h^3}, \qquad O(h).$$

$$f^{(4)}(x) \approx \frac{f(x) - 4f(x-h) + 6f(x-2h) - 4f(x-3h) + f(x-4h)}{h^4}, \qquad O(h).$$

$$f^{(5)}(x) \approx \frac{f(x) - 5f(x-h) + 10f(x-2h) - 10f(x-3h) + 5f(x-4h) - f(x-5h)}{h^5}, \qquad O(h).$$

### 2.1.3 Central differences

$$f''(x) \approx \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}, \qquad O(h^2).$$

$$f^{(3)}(x) \approx \frac{f(x+2h) - 2f(x+h) + 2f(x-h) - f(x-2h)}{2h^3}, \qquad O(h^2).$$

$$f^{(4)}(x) \approx \frac{f(x+2h) - 4f(x+h) + 6f(x) - 4f(x-h) + f(x-2h)}{h^4}, \qquad O(h^2).$$

$$f^{(5)}(x) \approx \frac{f(x+3h) - 4f(x+2h) + 5f(x+h) - 5f(x-h) + 4f(x-2h) - f(x-3h)}{2h^5}, \qquad O(h^2).$$

The main conclusion is that the central formulas are more accurate for the same step size because their leading truncation error is $O(h^2)$ rather than $O(h)$.

### 2.2 Problem 4(c): error model and optimal step size

For the forward-difference definition

$$f'(x) \approx \frac{f(x+h) - f(x)}{h},$$

the total error is the sum of truncation and roundoff contributions. For $f(x) = \sin x$ at $x = 0.5$, the lecture gives the model

$$g(h) = \left| \frac{h}{2} \sin(0.5) + \frac{\tilde{\varepsilon}}{h} \right|, \qquad \tilde{\varepsilon} = 7 \times 10^{-17}.$$

Balancing the two terms gives

$$\frac{1}{2} \sin(0.5) - \frac{\tilde{\varepsilon}}{h^2} = 0,$$

so the predicted best step is

$$h_{\text{best}} = \sqrt{\frac{2\tilde{\varepsilon}}{\sin(0.5)}} \approx 1.71 \times 10^{-8}.$$

This value should be close to the minimum seen in the numerical log-log plot.

## 3 Numerical Results

### 3.1 Problem 2: approximating $f'(2)$ for $f(x) = xe^x$

The exact derivative is

$$f'(x) = (x+1)e^x, \qquad f'(2) = 3e^2 \approx 22.1671682968.$$

Using the table values with $h = 0.1$, the results are shown in Table 2.

表 2: Problem 2 results from the given table data at $h = 0.1$.

| Method | Approximation | Absolute error |
|---|---|---|
| Forward difference | 23.70845000 | 1.54128170 |
| Backward difference | 20.74913000 | 1.41803830 |
| Central difference | 22.22879000 | 0.06162170 |
| Forward 3-point | 22.03231000 | 0.13485830 |
| Backward 3-point | 22.05452500 | 0.11264330 |
| Central 5-point | 22.16699917 | 0.00016913 |

The central 5-point formula is clearly the best one for the tabulated data. When the step size is swept from $10^{-1}$ to $10^{-12}$, the smallest reported error is $3.55 \times 10^{-12}$ for the central 5-point method at $h = 10^{-3}$. After that, the error increases again because subtractive cancellation and floating-point roundoff begin to dominate.

## 3.2 Problem 3: approximating $f''(2)$ for $f(x) = xe^{x^2}$

For this function,

$$f''(x) = (6x + 4x^3)e^{x^2}, \qquad f''(2) = 44e^4 \approx 2402.3186014583.$$

The table-data results are listed in Table 3.

表 3: Problem 3 results from the given table data at $h = 0.1$.

| Method | Approximation | Absolute error |
|---|---|---|
| Forward 3-point | 4189.712800 | 1787.394199 |
| Backward 3-point | 1468.599900 | 933.718701 |
| Central 3-point | 2460.877300 | 58.558699 |
| Central 5-point | 2399.497458 | 2.821143 |

Again the central 5-point formula performs best. In the $h$-sweep, its smallest reported error is $6.37 \times 10^{-8}$ at $h = 10^{-3}$. When $h$ is reduced below about $10^{-4}$, the roundoff term grows rapidly because the formula divides by $h^2$, so very small perturbations in the function values are strongly amplified.

## 3.3 Problem 4(a) and 4(b): forward-difference derivative of $f(x) = x(x-1)$

The analytical derivative is

$$f'(x) = 2x - 1, \qquad f'(1) = 1.$$

For part (a), the forward-difference program with $\delta = 10^{-2}$ gives

$$f'(1) \approx 1.01,$$

so the absolute error is exactly $0.01$.

For part (b), the second program evaluates a sequence of $\delta$ values. The results are summarized in Table 4.

表 4: Problem 4(b): derivative accuracy versus $\delta$.

| $\delta$ | Numerical derivative | Absolute error |
|---|---|---|
| $10^{-2}$ | 1.0100000000000009 | $1.000000 \times 10^{-2}$ |
| $10^{-4}$ | 1.0000999999998899 | $1.000000 \times 10^{-4}$ |
| $10^{-6}$ | 1.0000009999177333 | $9.999177 \times 10^{-7}$ |
| $10^{-8}$ | 1.0000000039225287 | $3.922529 \times 10^{-9}$ |
| $10^{-10}$ | 1.0000000828403710 | $8.284037 \times 10^{-8}$ |
| $10^{-12}$ | 1.0000889005833413 | $8.890058 \times 10^{-5}$ |
| $10^{-14}$ | 0.9992007221626509 | $7.992778 \times 10^{-4}$ |
| $10^{-16}$ | 0.0000000000000000 | $1.000000 \times 10^{0}$ |
| $10^{-18}$ | 0.0000000000000000 | $1.000000 \times 10^{0}$ |

The best value in this list is $\delta = 10^{-8}$. The data show the standard pattern very clearly: decreasing $\delta$ first reduces truncation error, but eventually the finite precision of the machine destroys the subtraction $f(x+\delta) - f(x)$.

## 3.4 Problem 4(c): reproducing the lecture error plot

The script `hw7_4/c/c.py` reproduces the lecture plot for the forward-difference approximation of $\sin x$ at $x = 0.5$. The numerical sweep uses

$$h = 10^{-0.1i}, \qquad i = 0, 1, \ldots, 200.$$

The script reports the best sampled value

$$h_{\text{best,num}} = 10^{-8},$$

which is consistent with the theoretical estimate

$$h_{\text{best,theory}} \approx 1.71 \times 10^{-8}.$$

The agreement is good because the sampled grid in $\log_{10}(h)$ has spacing $0.1$, so $10^{-8}$ is the nearest tested point to the theoretical optimum.

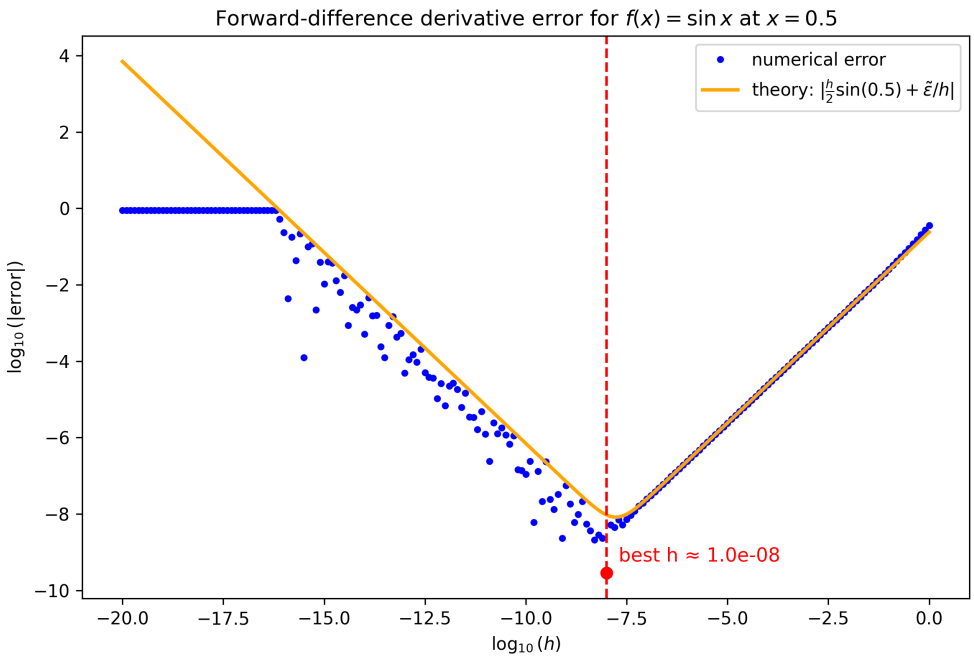


图 1: Problem 4(c): numerical forward-difference error and the lecture error model for $f(x) = \sin x$ at $x = 0.5$.

# 4 Discussion

The four tasks illustrate the same practical lesson from different angles. Higher-order symmetric formulas usually outperform one-sided formulas when the same data are available, because their truncation error is smaller. This is why the central 5-point formulas are the best performers in both Problems 2 and 3.

At the same time, the homework also shows why "smaller step size is always better" is false. When $h$ or $\delta$ becomes too small, the subtraction of nearly equal floating-point numbers loses significant digits, and division by $h$ or $h^2$ magnifies that loss. The result is an optimal intermediate step size rather than a monotonic improvement.

The selected source files are internally consistent with this interpretation. The electronic derivation in Problem 1, the complete Python workflows in Problems 2 and 3, and the direct lecture-matching scripts in Problem 4 together form a coherent final submission set.

## 4.1 Conclusion

The final conclusions of Homework 7 are:

(1) Central finite-difference formulas are more accurate than forward or backward formulas at the same step size when symmetric data are available.

(2) For Problem 2 and Problem 3, the central 5-point formulas are the most accurate among the tested methods.

(3) Reducing $h$ improves accuracy only up to a point; after that, roundoff error dominates.

(4) In Problem 4(c), the theoretical optimum $h_{\text{best}} \approx 1.71 \times 10^{-8}$ agrees well with the numerical minimum near $10^{-8}$.

*End of Report*